Constellation
Clinton Power Station
8401 Power Road
Clinton, IL 61727

U- 604869 10 CFR 50.73
March 3, 2026

U.S. Nuclear Regulatory Commission
ATTN: Document Control Desk
Washington, D.C. 20555-0001

Clinton Power Station, Unit 1
Renewed Facility Operating License No. NPF-62
NRC Docket No. 50-461

Subject: Supplemental Licensee Event Report 2025-002-01

Enclosed is Supplemental Licensee Event Report (LER) 2025-002-01: Automatic SCRAM on High Reactor Water Level. This report is being submitted in accordance with the requirements of 10 CFR 50.73.

There are no regulatory commitments contained in this report.

Should you have any questions concerning this report, please contact Mr. Derek Hillinger, Regulatory Assurance Manager, at (217) 937-2800.

Respectfully,

Frank Payne
Site Vice President
Clinton Power Station

Attachment: Licensee Event Report 2025-002-01

cc:

Regional Administrator - Region III
NRC Senior Resident Inspector - Clinton Power Station
Office of Nuclear Facility Safety - Illinois Emergency Management Agency

NRC FORM 366 U.S. NUCLEAR REGULATORY COMMISSION
(04-02-2024)

APPROVED BY OMB: NO. 3150-0104 EXPIRES: 04/30/2027

LICENSEE EVENT REPORT (LER)

(See Page 2 for required number of digits/characters for each block)

(See NUREG-1022, R.3 for instruction and guidance for completing this form http://www.nrc.gov/reading-rm/doc-collections/nuregs/staff/sr1022/r3/)

1. Facility Name		2. Docket Number	3. Page
Clinton Power Station	☒ 050 / ☐ 052	00461	1 OF 2

4. Title
Automatic SCRAM on High Reactor Water Level

5. Event Date			6. LER Number			7. Report Date		
Month	Day	Year	Year	Sequential Number	Revision No.	Month	Day	Year
10	06	2025	2025	- 002 -	01	12	06	2025

8. Other Facilities Involved

Facility Name		Docket Number
n/a	☐ 050	n/a
n/a	☐ 052	n/a

9. Operating Mode: Mode 1
10. Power Level: 23

11. This Report is Submitted Pursuant to the Requirements of 10 CFR §: *(Check all that apply)*

10 CFR Part 20	☐ 20.2203(a)(2)(vi)	**10 CFR Part 50**	☐ 50.73(a)(2)(ii)(A)	☐ 50.73(a)(2)(viii)(A)	☐ 73.1200(a)
☐ 20.2201(b)	☐ 20.2203(a)(3)(i)	☐ 50.36(c)(1)(i)(A)	☐ 50.73(a)(2)(ii)(B)	☐ 50.73(a)(2)(viii)(B)	☐ 73.1200(b)
☐ 20.2201(d)	☐ 20.2203(a)(3)(ii)	☐ 50.36(c)(1)(ii)(A)	☐ 50.73(a)(2)(iii)	☐ 50.73(a)(2)(ix)(A)	☐ 73.1200(c)
☐ 20.2203(a)(1)	☐ 20.2203(a)(4)	☐ 50.36(c)(2)	☒ 50.73(a)(2)(iv)(A)	☐ 50.73(a)(2)(x)	☐ 73.1200(d)
☐ 20.2203(a)(2)(i)	**10 CFR Part 21**	☐ 50.46(a)(3)(ii)	☐ 50.73(a)(2)(v)(A)	**10 CFR Part 73**	☐ 73.1200(e)
☐ 20.2203(a)(2)(ii)	☐ 21.2(c)	☐ 50.69(g)	☐ 50.73(a)(2)(v)(B)	☐ 73.77(a)(1)	☐ 73.1200(f)
☐ 20.2203(a)(2)(iii)		☐ 50.73(a)(2)(i)(A)	☐ 50.73(a)(2)(v)(C)	☐ 73.77(a)(2)(i)	☐ 73.1200(g)
☐ 20.2203(a)(2)(iv)		☐ 50.73(a)(2)(i)(B)	☐ 50.73(a)(2)(v)(D)	☐ 73.77(a)(2)(ii)	☐ 73.1200(h)
☐ 20.2203(a)(2)(v)		☐ 50.73(a)(2)(i)(C)	☐ 50.73(a)(2)(vii)		

☐ **OTHER** (Specify here, in abstract, or NRC 366A).

12. Licensee Contact for this LER

Licensee Contact	Phone Number (Include area code)
Derek Hillinger, Regulatory Assurance Manager	(217) 937-2800

13. Complete One Line for each Component Failure Described in this Report

Cause	System	Component	Manufacturer	Reportable to IRIS	Cause	System	Component	Manufacturer	Reportable to IRIS
B	SJ	P	G080	Y					

14. Supplemental Report Expected		15. Expected Submission Date	Month	Day	Year
☐ No	☐ Yes (If yes, complete 15. Expected Submission Date)				

16. Abstract (Limit to 1326 spaces, i.e., approximately 13 single-spaced typewritten lines)

On October 6, 2025, during reactor startup, the Clinton Unit 1 reactor automatically scrammed due to high reactor water level.

The cause of the high reactor water level was a failure of a Turbine Driven Reactor Feed Pump (TDRFP) servo valve that resulted in excessive feedwater flow to the reactor vessel.

This event is reportable in accordance with 10 CFR 50.73(a)(2)(iv)(A) for “Any event or condition that resulted in manual or automatic actuation of any of the systems listed in paragraph (a)(2)(iv)(B) of this section.” The specific system listed in that paragraph that actuated was the “Reactor protection system (RPS) including: reactor scram or reactor trip.”

1. FACILITY NAME		2. DOCKET NUMBER	3. LER NUMBER		
			YEAR	SEQUENTIAL NUMBER	REV NO.
Clinton Power Station	■ 050 / ☐ 052	00461	2025	- 002	- 00

NARRATIVE

CONDITION PRIOR TO THE EVENT

Unit: 1 Event Date: October 6, 2025 Event Time: 03:21 CST
Mode: 1 Mode Name: Power Operation Reactor Power: 23%

EVENT DESCRIPTION

On October 6, 2025, during reactor startup from the Clinton Unit 1 refueling outage, Turbine Driven Reactor Feed Pump (TDRFP) ‘A ‘was placed in service. While manually raising the speed of the TDRFP, the operator noticed that there was no rise in speed initially. The operator attempted again to raise pump speed. Speed then rose rapidly to 1300 rpm and went above the target setpoint of 1000 rpm. The operator placed the pump speed control mode in automatic to lower speed. TDRFP speed stabilized for about a minute before speed unexpectedly began rising rapidly to 4300 rpm. The operator then tripped the TDRFP. At the time of the trip, reactor water level was approximately 47 inches and rising. Reactor water level continued to rise, and when it reached 52 inches, an automatic reactor scram occurred.

CAUSE OF THE EVENT

A Root Cause investigation determined the event was caused by scoring on the A-port side of the servo valve body, confirming abrasive interaction. The scoring, caused by lodged debris, increased friction or temporarily locked the servo swing plate. The servo motor had to overcome additional drag, which it could not do effectively. As a result, the valve did not respond to a decreasing demand signal, causing failure to control the operating cylinder responsible for closing the low-pressure steam control valve resulting in a Reactor Protective System (RPS) SCRAM on level 8.

SAFETY CONSEQUENCES

There were no safety consequences impacting plant or public safety as a result of this event. The reactor trip system responded automatically due to the trip signal received. Additionally, the TDRFP was tripped prior to the scram, so no additional feedwater would have been injected. There was no loss of any function that would have prevented fulfillment of actions necessary to 1) shutdown the reactor and maintain it in a safe shutdown condition, 2) remove residual heat, 3) control the release of radioactive material, or 4) mitigate the consequences of an accident
There was no loss of safety function for this event.

CORRECTIVE ACTIONS

TDRFP B has the same control loop as TDRFP A and would be susceptible to the same failures under the conditions that occurred during start-up. To mitigate the same failures, the TDRFP B servo valve SV-2 was replaced before operation. To reduce risk during future startups, operating procedures will be changed to exclusively utilize a “rolling standby” mode before feeding the Reactor Pressure Vessel. Servo tested at power labs and final report of findings issued back to CPS.

Prior data of historical start-ups that included perturbations of the feed pump show evidence of starting up and having potential material or air in the oil that works through the system during initial start-up.

PREVIOUS OCCURRENCES

A review of Licensee Event Reports for the past five years identified no previous similar occurrences at Clinton Power Station.

U-604869

SUBJECT: Supplemental Licensee Event Report 2025-002-01

bcc:

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